External Collapse-Ordering Invariance Assessment on Frozen HAOS-IIP Telemetry

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Abstract

This paper freezes release `53.1` as a narrow post-foundations validation instrument layered above frozen HAOS-IIP telemetry. It is not a new phase, not a modification of HAOS-IIP core, and not a positive collapse-law claim. The instrument treats HAOS telemetry as a fixed black-box sensor, builds a score-ordered edge filtration on a graph-like artifact, perturbs that construction across locality, coupling, curvature, noise, and rewiring schedules, and asks only one bounded question: does the observed collapse ordering survive perturbation strongly enough to count as an invariant structural signal rather than an artifact of construction? On the current deterministic three-sector toy graph, a permissive toy control sensor yields `alpha` > `beta` > `gamma`, flip rate 0.069, persistent gap behavior, and an invariant-signal classification. The HAOS-IIP-backed adapter, which reuses only frozen telemetry, instead yields `alpha = beta` > `gamma`, flip rate 0.092, vanishing gap behavior, and an artifact-of-construction classification. The bounded conclusion is therefore negative and useful: the external runner is now sufficient to separate protocol-level apparent signal from frozen-sensor structural support, but the present toy artifact does not yet earn a positive HAOS-IIP collapse-ordering claim.

1 Scope

This release sits closer to `45.1` and `50.3` than to a new phase paper.

`45.1` compressed frozen telemetry into a callable structural-stability oracle. `50.3` then defined the post-foundations discipline: validation should ask whether a candidate survives perturbation, purity checks, and scaling closure strongly enough to support bounded language. Release `53.1` applies that discipline to a new external question:

> if a collapse ordering appears on a graph artifact, does it survive perturbation as structure?

This paper does not introduce:

- a new HAOS-IIP operator;
- a new frozen telemetry channel;
- a new sector claim;
- a continuum or ontology claim;
- a positive collapse-ordering law.

It does introduce:

- an external invariance-assessment protocol layered above frozen telemetry;
- a strict distinction between protocol output and structural support;
- a current negative result on the tested toy artifact under the HAOS-IIP-backed sensor.

2 External-Layer Rule

The architecture is intentionally separated:

- HAOS-IIP core remains frozen;
- frozen telemetry remains the only authority sensor;
- the validation runner is external and swappable;
- the result is whatever survives perturbation under the frozen sensor.

The runner therefore treats HAOS metrics only as black-box observables. It does not reinterpret, extend, or reopen the frozen core.

3 Protocol

3.1 Inputs

Each run consumes:

- a graph $G = (V, E)$ with sector labels on nodes;
- edge attributes $\mathrm{strength}(i, j)$ and distance $d(i, j)$;
- locality parameter λ;
- sector-coupling weights;
- a curvature-penalty schedule;
- a black-box HAOS sensor returning

  $$\{\Delta\mathrm{persistence},\ k_*,\ \mathrm{safety\_margin},\ \mathrm{confidence}\}$$

  per inclusion step.

3.2 Filtration

Edges are scored by

$$\mathrm{score}(i, j) = \mathrm{strength}(i, j)\ \exp\left(-\frac{d(i, j)}{\lambda}\right)\ \mathrm{curvature\_penalty}(i, j).$$

The runner sorts edges by descending score and incrementally includes them to form a filtration G_t. After each inclusion step it calls the black-box sensor and records the resulting time series.

3.3 Sweeps and perturbations

The current deterministic schedule contains 36 runs across:

- locality schedules with coarse, baseline, and fine λ steps;
- sector-coupling rescalings;
- curvature-penalty rescalings;
- weight noise in the $\pm5\%$ to $\pm10\%$ range;
- bounded random reshuffling on roughly 10% of the edges.

3.4 Strict outputs

The runner emits only:

- collapse ordering by sector;
- consistency across trials, reported as pairwise flip rate;
- gap behavior, reported as `persists`, `shrinks`, or `vanishes`;
- one bounded assessment: invariant structural signal or artifact of construction.

No interpretation beyond invariance assessment is part of the output contract.

4 Current Deterministic Artifact

The tested artifact is a small three-sector graph with sectors `alpha`, `beta`, and `gamma`. Its role is deliberately modest:

- it is large enough to produce nontrivial filtration orderings;
- it is small enough for deterministic perturbation scans;
- it is not treated as a physical model.

Two sensor backends were applied to the same artifact:

- a toy control sensor, used only as a protocol sanity check;
- a HAOS-IIP-backed sensor adapter, built only from frozen telemetry.

The toy sensor is not authoritative. Its job is to show that the runner can register a robust ordering when the backend is permissive. The HAOS-IIP-backed adapter is the meaningful read because it uses the frozen telemetry layer already accepted by the repository.

5 Frozen Quantitative Summary

The important row is the second one. Under the frozen telemetry adapter, the apparent ordering does not survive as a clean ordered ladder. The first two sectors hit k_* simultaneously, and the adjacent gap does not survive perturbation strongly enough to support a structural-ordering claim.

backend	collapse ordering	k_* steps	flip rate	gap behavior	assessment
toy control	`alpha > beta > gamma`	`11, 12, 15`	0.069	`persists`	`invariant structural signal`
HAOS-IIP-backed	`alpha = beta > gamma`	`10, 10, 12`	0.092	`vanishes`	`artifact of construction`

Table 1: Current deterministic invariance-assessment read on the three-sector toy artifact.

6 What 53.1 Now Supports

The strongest honest statement supported by this release is:

> frozen HAOS-IIP telemetry can now be used as a black-box invariance-assessment sensor inside an external validation runner, and on the present toy artifact that runner rejects a positive collapse-ordering claim.

That is not a failure of the instrument. It is the first useful property the instrument should have. A validation runner is more valuable when it can kill an apparent signal honestly than when it can always recover one.

7 What 53.1 Refuses to Say

This release does not justify any of the following:

- that collapse ordering is already a validated HAOS-IIP observable;
- that `alpha > beta > gamma` is a structural law;
- that the current toy graph is more than a probe artifact;
- that the toy control sensor should be treated as evidence;
- that any geometry-like, force-like, or continuum-like interpretation is now earned.

The negative HAOS-IIP-backed outcome is therefore part of the result, not an obstacle to it.

8 Why This Matters

Without this runner, one could easily mistake a construction-dependent ordering for a structural signal. The present release closes that loophole in a bounded way:

- the core remains frozen;
- the probe layer is external;
- the backend can be swapped;
- only invariant survival under the frozen backend counts.

That is exactly the kind of discipline a post-foundations validation program should add before stronger language is allowed.

9 Right Next Move

The next practical step is not to upgrade the claim. It is to improve the artifact class:

- test graph families beyond the present toy artifact;
- introduce sector layouts whose ties are not built into the construction;
- keep the same runner and the same frozen sensor interface;
- accept a positive statement only if the HAOS-IIP-backed path, not the toy control path, preserves the gap.

In other words, the current runner should now be reused, not reinterpreted.

10 Conclusion

Release `53.1` freezes an external collapse-ordering invariance-assessment protocol on frozen HAOS-IIP telemetry. The protocol is intentionally narrow, black-box, and perturbation-first. On the current deterministic toy artifact, the permissive control backend produces an apparent ordered signal, but the HAOS-IIP-backed backend classifies the same artifact as construction-sensitive because the leading gap vanishes under perturbation and the first two sectors collapse together. The bounded conclusion is therefore methodological and negative: the validation runner is now useful, but no positive HAOS-IIP collapse-ordering structural signal is yet earned from the present artifact.